

Problem Set 1

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1. Sleep

```
ramisk@Ramis-Laptop:~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_
ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forp
han user/_dorphan user/_sleep user/_sixfive user/_memdump user/_find user/_uptime user/_history user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1196 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -
drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sleep 10
$ |
```

2. Sixfive

```
ramisk@Ramis-Laptop: ~/xv6 X + v
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -
drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sleep 10
$ sixfive sixfive.txt
5
100
18
6
```

3. Memdump

```
6
$ memdump
Example 1:
61810
2025
Example 2:
a string
Example 3:
another
Example 4:
BB0
1819438967
100
z
xyzy
Example 5:
hello
w
o
r
```

4. Find

```
ramisk@Ramis-Laptop: ~/xv6-labs-2025$ make qemu
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mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_
ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forp
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drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
$ |
```

5. Exec

```
ramisk@Ramis-Laptop: ~/xv6-labs-2025$ make qemu
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mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_
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drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sh < findtest.sh
hello
hello
hello
$ |
```

Make Grade

```
ramisk@Ramis-Laptop: ~/xv6
== Test memdump, format ii, S, p ==
$ make qemu-gdb
memdump, format ii, S, p: OK (1.4s)
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (2.3s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (2.7s)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (3.0s)
== Test exec ==
$ make qemu-gdb
exec: OK (2.0s)
== Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: OK (1.8s)
== Test exec, recursive find ==
$ make qemu-gdb

Timeout! exec, recursive find: OK (31.7s)
== Test time ==
time: FAIL
    Cannot read time.txt
Score: 130/131
```